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EXERCISE DEVICE HAVING DYNAMIC LOAD TRANSFER SYSTEM

Abstract

A climbing exercise device integrates a dynamic load transfer system, optional rebound mechanism, and optional flywheel or magnetic brake to deliver fluid, full-body workouts while reducing knee strain. Opposing foot carriages, each guided along an upright support, are linked by flexible webbing wrapped around a rotatable transfer shaft. As one foot carriage descends and unwinds its webbing, the other carriage is lifted, shifting the load ratio to ease knee stress at deeper flexion. A rebound mechanism keeps the webbing under tension and increases that tension at stroke limits, promoting quick, smooth directional changes. An optional flywheel or magnetic brake further refines motion by adding adjustable resistance and inertia without impeding rapid reversals. Arm handles may be coupled—directly or cross-linked—to the foot carriages for upper-body engagement, resulting in a balanced, efficient climb. Multiple embodiments and frame configurations accommodate various height preferences, storage options, and user needs while retaining the synergy of these key systems.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Patent Application No. 63/562,408, filed on Mar. 7, 2024, U.S. Provisional Patent Application No. 63/637,001, filed Apr. 22, 2024 and U.S. Provisional Patent Application No. 63/681,998, filed Aug. 12, 2024, the disclosures of which are incorporated by reference herein in their entireties.

FIELD OF THE INVENTION

[0002] The present invention relates to exercise equipment and, more particularly, to a climbing exercise device configured to simulate vertical climbing movements by utilizing a novel synergistic combination of systems.

BACKGROUND OF THE INVENTION

[0003] Conventional climbing exercise machines generally confine users to a limited range of motion. Typically, these devices require simultaneous arm and leg actions, where the user grips handlebars while standing on foot pedals that travel along a fixed vertical path-often restricting each limb's movement to no more than 21 inches.

[0004] Such machines can be broadly classified as follows: [0005] Ladder Simulators: Devices that mimic ladder climbing, with the hand and foot on the same side moving concurrently in the same direction-a motion commonly referred to as the “standard pattern.” [0006] Rock Climbing Simulators (Vertical Crawling or Contra-Lateral): Systems designed to imitate rock climbing by allowing hand-to-foot movements similar to crawling. In this mode, the hand and foot on the same side move closer together as the knee bends and then diverge as the leg extends.

[0007] Some hybrid machines blend these patterns to offer greater versatility.

[0008] A common drawback in all of these types of climbing exercise machines is the high amount of strain placed on the user's legs, especially on the user's upper leg during its downward stroke from a bent-knee position. This is because the user needs to overcome their own body weight on the opposite, lower leg by pushing down with the upper leg. This pushing force, which becomes more significant with a deeper knee bend, can lead to user discomfort and/or injury, particularly to the user's knees.

[0009] Another drawback in these types of climbing exercise machines is that the transition from upstroke to downstroke, and vice versa, can occur abruptly, thereby impeding fluid motion and/or reducing user comfort.

[0010] Such drawbacks may discourage a user from incorporating climbing exercise machines into their workouts.

SUMMARY OF THE INVENTION

[0011] It is an object of the present invention to provide a climbing exercise device that allows a user to simulate vertical climbing movements while avoiding one or more of the drawbacks noted

above. Embodiments of the climbing exercise device may achieve this object by use of a dynamic load transfer system that continuously redistributes the user's weight throughout the climbing motion, reducing strain on the lower body and promoting a smoother, more efficient exercise. [0012] In a preferred embodiment, the dynamic load transfer system comprises webbing segments and a rotatable shaft arranged to continuously redistribute a user's weight throughout the climbing motion via a controlled winding and unwinding of the respective webbing segments around the rotating shaft in an alternating manner. Webbing segments may extend from the rotatable shaft to respective foot carriages on opposite sides of the exercise device such that, e.g., the rotatable shaft can be rotated in a first direction in response to downward movement of a first of the foot carriages, and rotated in a second direction (preferably opposite the first direction) in response to downward movement of a second of the foot carriages. During an upstroke of a respective foot carriage, the webbing segment associated with that foot carriage is caused to wrap around the rotating shaft, increasing the distance between the axis of rotation of the shaft and any downward force exerted by the respective foot carriage via the webbing segment. As such, when a downward force is applied to the respective foot carriage at the top of its travel, a torque, moment, or mechanical advantage on that side of the exercise device is noticeably increased relative to a bottom of its travel. This simple, cost-effective dynamic load transfer system noticeably reduces strain during the downward stroke and promotes an efficient, natural movement.

[0013] In some embodiments, the dynamic load transfer system may be combined with a rebound mechanism that provides a controlled and consistent rebound at both the top and bottom of each stroke. In an example embodiment, the rebound mechanism comprises an elastic cable assembly (such as a flexible cable with one or more tension springs disposed along its length to provide elasticity) connected between right and left foot carriages via a pulley. The elastic cable assembly is configured such that tension in the cable assembly is at a minimum when the foot carriages are at the same height, and the tension in the cable assembly increases as the difference in height of the foot carriages increases (e.g., as one foot carriage moves up and the other foot carriage moves down). Such a rebound mechanism is easily integrated with the dynamic load transfer system and can help facilitate quick changes in direction without abruptness and enhance overall motion fluidity, contributing to a more enjoyable climbing experience.

[0014] In some embodiments, the climbing exercise device may further comprise an optional resistance system, such as a friction brake acting on the rotatable shaft, to provide adjustable resistance levels. In some embodiments, a flywheel may be coupled with the rotating shaft to smooth out the movement without impeding on the transitions. When a flywheel is provided, a resistance system (such as a friction brake) may be configured to act on the flywheel, ensuring that the workout remains both challenging and controlled without introducing unnecessary complexity, all while maintaining the natural cadence of the climbing motion.

[0015] Traditional vertical climbers often induce early fatigue unless the user shortens the stroke length; by contrast, this device allows for a deeper range of motion and extended workout sessions. In one embodiment, two independent foot pedals—each mounted on its own guide carriage—are linked so that when one descends, the other rises.

[0016] In the first illustrated embodiment, a guide rail column extends from the floor base to above the user's head; in alternative embodiments (the second through fifth illustrated embodiments), the rail terminates near the user's hips. In all configurations, the design emphasizes biomechanical efficiency and smooth motion.

[0017] Furthermore, the dynamic load transfer system not only reduces knee strain but can also be configured to incorporate arm workouts. In the first illustrated embodiment, movable hand carriages or arm handle assemblies are provided to facilitate arm workouts. In this embodiment, the arm resistance is higher at the start of the downward pull—matching the user's stronger upper stroke position—and tapers off toward the lower end for a balanced, full-body workout. Additionally, cords in this embodiment link the foot carriages to corresponding hand carriages,

introducing a coordinated change in arm pulling force that complements the load transfer effect.

[0018] In the second through fifth illustrated embodiments, each arm handle assembly is coupled to the opposite foot carriage via a hidden crossover mechanism positioned generally below the user's knee elevation. This design contrasts with prior art hybrid climbers that employ crossover handles connecting at the arm handle level, which can clutter the area near the user's eye level. The concealed linkage not only maintains aesthetic elegance but also ensures that the arm pulling force directly matches the foot's shifting load. A mechanical advantage in the arm assembly further assists the user in lowering the opposite foot, further promoting a fluid and less demanding climb.

[0019] All of the illustrated embodiments produce synchronized movement that engages both the upper and lower body, resulting in a comprehensive workout.

[0020] Together, these synergistic systems provide a climbing exercise device that not only overcomes one or more of the inherent limitations of conventional designs but also offers a uniquely advantageous, cost-effective, and user-friendly exercise experience that minimizes lower-body strain and maximizes user comfort.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] Embodiments of the present invention will now be described, by way of example only, with reference to the following drawings, in which:

[0022] FIG. 1 is a rear-side perspective view of a first embodiment of the exercise device.

[0023] FIG. 2 is a front-left perspective view of the exercise device of FIG. 1.

[0024] FIG. 3 is a partial rear perspective view of the lower portion of the exercise device of FIG. 1, showing the right foot roller assembly in its lowest position while the left foot roller assembly is in its highest position.

[0025] FIG. 4 is the same partial rear perspective view as FIG. 3, with the foot roller assemblies reversed relative to FIG. 3.

[0026] FIG. 5 is a partial front-right perspective view of the lower portion of the exercise device of FIG. 1, further illustrating the foot roller assembly and a foot-to-foot cable assembly located beneath it.

[0027] FIG. 6 is a partial right-side perspective view of the lower portion of the exercise device of FIG. 1.

[0028] FIG. 7 is a mostly rear-right perspective view of the base of the exercise device of FIG. 1, showing the rear base kickstand.

[0029] FIG. 8 is the same view shown in FIG. 7, illustrating a movable gusset on the rear kickstand as positioned before or after folding the kickstand for storage or roll-away transport.

[0030] FIG. 9 is the same view shown in FIGS. 7 and 8, with the kickstand fully stowed.

[0031] FIG. 10 is a left-side perspective view of the exercise device of FIG. 1 in a folded state for storage or transport.

[0032] FIG. 11 is a slightly rotated rear-right perspective view of the exercise device of FIG. 1 in the same folded state shown in FIG. 10.

[0033] FIG. 12 is a left-side perspective view of a second embodiment of the exercise device.

[0034] FIG. 13 is a rear-right perspective view of the exercise device of FIG. 12.

[0035] FIG. 14 is a rear view of the exercise device of FIG. 12, including a cutaway portion.

[0036] FIG. 14A is an enlarged, cutaway view of the portion circled in phantom line in FIG. 14, illustrating how the arm poles couple to the foot carriages.

[0037] FIG. 15 shows the foot carriages of the exercise device of FIG. 12, isolated for clarity.

[0038] FIG. 16 is a top-down rear-right perspective view of the second embodiment in a folded state.

[0039] FIG. **17** is a partial front-right perspective view of the lower portion of the exercise device of FIG. **12**.

[0040] FIG. **18** is a rear-right perspective view of the second embodiment in its folded state.

[0041] FIG. **19** is a left-side perspective view of a third embodiment of the present invention.

[0042] FIG. **20** is a bottom view of the third embodiment of FIG. **19**.

[0043] FIG. **21** is a mostly front-left perspective view of the third embodiment.

[0044] FIG. **22** is a rear-left perspective view of the third embodiment in a stowed position.

[0045] FIG. **23** is a front-left perspective view of a fourth embodiment.

[0046] FIG. **24** is a rear-right perspective view of the fourth embodiment.

[0047] FIG. **25** is another rear-right perspective view of the fourth embodiment.

[0048] FIG. **26** is the same view of the fourth embodiment shown in FIG. **25**, with certain components removed to more clearly illustrate the dynamic load transfer and dynamic rebound system.

[0049] FIG. **27** is a right-front perspective view of a fifth embodiment.

[0050] FIG. **27A** is an enlarged, sectioned view of the area circled in phantom line in FIG. **27**.

[0051] FIG. **28** is a rear-left perspective view of the fifth embodiment in a stowed position.

DETAILED DESCRIPTION

[0052] Referring to the accompanying illustrative drawings, FIGS. **1-11**, and particularly FIGS. **1** and **2**, the exercise device **100** according to a first embodiment includes an upright guide rail column **110**. This column (also referred to herein as an upright support column) is supported in a generally upright and slightly forward-tilted position by its lower end, which is bolted to a base, shown as a metal base weldment **120**, configured to rest on a floor surface. The base weldment includes a horizontal base tube **122**, with a mid-point upright support **128** extending upward parallel to the column **110**. The upright support **128** is fastened midway along the column's height via a welded bridge tube **114** and mounting plate with bolt-through holes. The device also features a pivoting front kickstand **116**, which is attached to the base weldment's upper end by a pivot bolt **117**. A spring pin and hole mechanism provides a means to hold the kickstand securely in either its outward or folded inward position.

[0053] The guide rail column **110** is equipped with full-length, concave-profiled channels **110g** along both its front and back sides, as depicted in FIG. **2**. These channels interact with multiple rollers **195t** mounted on foot carriages **130** (**130r** on the right, and **130s** on the left), ensuring smooth and stable movement. Each foot roller carriage **130** is equipped with four half-round-faced rollers **195t**, similar to rollerblade wheels, and an additional forward roller **197**. The forward roller **197** engages with the left and right walls of the upright support **128**, contributing to structural stability and alignment of the foot carriages **130**. Similarly, each arm roller carriage **140r** and **140s** includes four rollers **195t**, which glide within the concave-profiled channels **110g**, ensuring consistent and smooth operation.

[0054] A dynamic load transfer system, which is central to this invention, includes a transfer shaft **151** rotatably mounted on the column **110** and webbing segments **152** (**152r** on the right, and **152s** on the left) attached between foot carriages **130** and transfer shaft **151**. Transfer shaft **151** is shown as a single shaft extending through the column **110**, but it will be appreciated that the transfer shaft may include more than one shaft, e.g., a first shaft on one side of the column and a second shaft on an opposite side of the column. In some embodiments, transfer shaft is rotatable about its longitudinal axis (also referred to herein as its axis of rotation). Each leg carriage **130** incorporates a webbing attachment weldment stud **133**, serving as an anchor point for webbing segments **152**. In the illustrated embodiment, webbing segments **152** are shown as flexible straps or belts. However, it will be appreciated that webbing segments may comprise other types of flexible members, such as cables and cords. In some embodiments, guides may be provided on the transfer shaft (or on another part of the exercise device) to limit lateral movement of the webbing segments in order to facilitate winding of the webbing segments around the transfer shaft in a manner that increases the

distance of the unwrapped portion of the webbing segments from the longitudinal (rotational) axis of the transfer shaft to provide additional mechanical advantage (i.e., increase torque or moment) on the downstroke. The webbing segments attach to these studs and are then wrapped around the transfer shaft **151**. The upper end of each web segment **152** engages into a precisely dimensioned slot at either end of the transfer shaft **151**. These slots provide a snug fit, ensuring the webbing remains securely seated during repetitive operation. Additional measures, such as a roll pin through a perpendicular through-hole in the slot, may optionally be employed to further stabilize the webbing.

[0055] As shown in FIGS. **3** and **4**, when one foot carriage **130** is fully lowered, its corresponding load-support webbing **152** wraps around the transfer shaft **151** for about two full rotations. Conversely, the webbing **152** of the fully raised foot carriage **130** wraps around the same transfer shaft in the opposite direction for approximately seven revolutions. This configuration ensures synchronized movement and proper load distribution between the opposing carriages. The difference in wrap diameters creates a 2:1 mechanical advantage for the fully raised foot carriage, facilitating efficient upward and downward motion. When the foot carriages are at the same elevation, as depicted in FIG. **5**, the wraps are equal, resulting in a 1:1 load distribution at that brief moment of operation.

[0056] The arm carriages **140** (**140s**, **140r**) also play a key role in coordinating upper and lower body movements. Each arm carriage includes an offset stud weldment **147**, which serves as an attachment point for the cables **161**. These cables wrap around idler pulleys **162** positioned at the uppermost portion of the guide rail column **110**, as shown in FIG. **2**, and connect to corresponding leg carriages via lower attachment points. The cable and pulley system creates a “crawling” or contra-lateral movement pattern, where the right foot is synchronized with the right hand, and the left foot with the left hand.

[0057] It should also be noted that the upper cable systems **160** could be replaced with rigid linkage members that directly attach each arm carriage to its corresponding leg carriage for “ladder” or “standard” movements. Alternatively, a rigid linkage could be formed to connect the left foot carriage **130s** to the right arm carriage **140r** and the right foot carriage **130r** to the left arm carriage **140s** to maintain the “crawling” movement pattern. These alternatives provide additional flexibility in tailoring the device's movement patterns to suit different user preferences and workout objectives.

[0058] A brake drum **155** is affixed to the transfer shaft **151**, providing adjustable resistance during exercise. The drum may be coupled with a simple brake strap or an eddy current magnetic braking system, as described for later embodiments. This resistance system affects all four user limbs simultaneously, offering a customizable workout experience. It is worth noting that a flywheel may also be incorporated to smooth out the ride by leveraging its momentum. However, the flywheel's mass must be carefully calibrated to prevent interference with rapid directional changes enabled by the rebound system **180**.

[0059] The rebound system, detailed in FIGS. **5** and **6**, consists of a cable **181**, tension springs **183**, and an idler pulley **182**. This system applies a counteracting force to the webbing on the foot carriages, maintaining proper tension and preventing slack that might otherwise cause the webbing to unspool or fail to spool properly around the transfer shaft **151**. The rebound springs or alternative tension mechanisms are least extended when the foot carriages are at the same elevation, as shown in FIG. **5**, and most extended when the foot carriages reach their travel limits, as shown in FIG. **6**. This variability is a result of the changing spooling diameters of the webbing around the transfer shaft. The added tension at the stroke extremes generates a substantial rebound effect, facilitating quick directional changes and further promoting a fluid, continuous climbing experience.

[0060] To provide a more detailed explanation of the rear base kickstand **124**, refer to FIGS. **7-9**. FIG. **7** illustrates the kickstand **124** in its operational position, ensuring that the machine remains

stable and does not tip backward or wobble during use. Without the kickstand, the user's body weight while operating the machine would tend to tip it backward. The rear kickstand extension arm **126** is adequately long and features a threaded adjustable foot **129**, allowing users to balance the machine's load and ensure even distribution across all four contact points with the floor. To fold the rear kickstand **124** inward, the user simply removes the stop pin **113** from the stop pin hole **121**. This action allows the slidable gusset weldment **125** to disengage from the column **110**'s metal channel stop trap and slide rearward along the extension arm **126**. The kickstand can then be rotated into a holding position, wedged between the column **110** and one of the lowered foot carriage structural members **132**, keeping it folded securely even when the machine is tilted backward for transport.

[0061] The foregoing description highlights the innovative integration of the dynamic load transfer system, rebound system, and optional resistance and flywheel mechanisms. While secondary components like the offset stud weldments (**133**, **147**) and cable systems support the device's functionality, the seamless interplay of the primary systems defines the invention. Variations in materials, configurations, or additional features may be implemented without departing from the scope of the invention.

Second Embodiment of the Exercise Device (**200**)

[0062] Turning now to the second embodiment of the exercise device, referenced as **200** and depicted in FIGS. **12-18**, this design shares many features with the first embodiment **100** while introducing key distinctions. The reference item numbers (2nd and 3rd digits) for embodiment **200** and subsequent embodiments (**300**, **400**, and **500**) correspond to those used in describing embodiment **100**, aiding in comparison and understanding.

[0063] The second embodiment **200** retains the dynamic load transfer system **250** and the “climbing” motion simulation of embodiment **100** but introduces a significant variation in the height of the guide rail column **210**. Unlike the taller guide rail column **110** in embodiment **100**, the guide rail column **210** of embodiment **200** terminates at an elevation generally corresponding to the user's hips when standing on both foot pedals. This modification accommodates users seeking less knee strain, a more compact machine, and a smoother, more fluid experience. The shorter guide rail column allows the device to be substantially lighter, more economical, and more space-efficient while maintaining functionality comparable to taller climbing machines, such as embodiment **100** and other vertical climbers on the market.

[0064] Another key distinction in embodiment **200** is the design of the arm handle assemblies **240**, which are directly coupled to the opposite leg carriages **230**. This coupling eliminates the interconnected cable and pulley systems (**160** and **170**) used in embodiment **100**. This simplified design reduces mechanical complexity while ensuring synchronized movement of the upper and lower body muscles, enabling a seamless climbing motion.

[0065] As seen in FIGS. **12** and **13**, embodiments **100** and **200** share the same dynamic load transfer system (**150** and **250**, respectively) and nearly identical leg carriage assemblies (**130** and **230**, respectively), which travel vertically along their respective guide rail columns (**110** and **210**). Both embodiments are supported by similar metal base frame components, ensuring stability in operation or when stowed. However, the lower guide rail height of embodiment **200** introduces opportunities for streamlining, such as substituting lighter or less costly materials for the guide rail, as demonstrated in embodiment **300** (FIGS. **19-21**) and the metal tubular guide rails of embodiments **400** and **500**. While embodiments **100** and **200** use wooden or comparable guide rail materials, subsequent embodiments incorporate alternative materials and structural designs.

[0066] To better understand the coupling of the arm handle assemblies **240** to the leg carriages **230**, refer to FIGS. **14**, **14 A-A**, and **15**. Each arm handle assembly **240** includes a stationary or rotatable handle **246** mounted to the upper 90-degree offset branches of arm insert rods **244**, which telescope within arm tubes **242**. Users can adjust the handle height by tightening a knob screw **245**, as shown in FIG. **16**. At the lower end of each arm tube **242**, a square insert bar **241** is affixed, which fits into

a square socket **248** welded to the bridge leg-to-arm bracket weldment **237**. This arrangement couples the arm handles to the opposite-side leg carriages (e.g., right arm assembly **240r** is linked to left leg carriage **230s**). As the user pushes downward on the left leg carriage **230s** with their foot, the coupled right arm handle **240r** moves downward in unison, creating the “crawling” movement pattern. The same applies to the left arm handle **240s** and right leg carriage **230r**.

[0067] In this embodiment, the arm handles **240** can be easily decoupled from the leg carriages **230** by simply lifting the arm handles upward, removing the square insert bars **241** from the square sockets **248**. This feature enhances portability, allows for compact storage, and facilitates easier shipping. For applications requiring a more stable connection, the arm handles can be permanently or semi-permanently affixed to their respective leg carriages.

[0068] As with the first embodiment, this design can accommodate alternative movement patterns. For instance, the arm handles could be directly coupled to the corresponding leg carriages for a “ladder” style movement or configured to achieve a “crawling” pattern with bent hand bar crossed linkages, as prior art has shown. However, the design demonstrated in embodiments **200-500** simplifies the crossover mechanism, concealing it below the user's hips for an elegant and streamlined approach.

[0069] FIG. **17** illustrates the machine in a stowed position, with the bridge leg-to-arm bracket weldment **237r** passing through a side slot **212** in the column **210**. The slot accommodates the full operational travel range of the foot carriages, roughly 21 inches, while allowing compact stowage. This embodiment also includes a rebound system **280**, similar to the rebound system **180** in embodiment **100**. As shown in FIG. **17**, adjustable threaded eyebolts **287** allow users to fine-tune the tension on the extension springs **283**, increasing or decreasing the rebound effect. Future embodiments, such as **400** and **500**, will demonstrate alternative rebound systems using compression springs concealed within the upright support tube.

[0070] The arm handles **240** are guided by “U”-grooved rollers **243**, which are mounted at the top end of the guide rail column **210**. These rollers provide low-friction support as the arm poles **242** reciprocate during exercise. Although plastic bushings or other low-cost alternatives could be used in place of rollers, the hard rubberized rollers **243** and rollerblade-style wheels **295t** ensure quiet and smooth operation, making this embodiment particularly appealing for residential or commercial use.

[0071] To enhance the workout experience, a flywheel and resistance system **250** is incorporated. The aluminum flywheel **255** is affixed to the transfer shaft **251** and interacts with a pair of magnets **253** that are pivotally adjustable via a threaded knob **257**. Turning the knob **257** lowers the magnets, increasing resistance as they surround the flywheel, or raises them to reduce or eliminate resistance. This system, best shown in FIG. **16**, provides customizable resistance while smoothing the ride. The flywheel also adds momentum, promoting a fluid and consistent motion without compromising the quick directional changes facilitated by the rebound system **280**. The inherent design of the dynamic load transfer system, where resistance decreases at the top of the stroke and increases at the bottom, further enhances the fluidity of the motion.

[0072] In summary, embodiment **200** introduces a more compact, streamlined design while maintaining the innovative features of embodiment **100**. The shorter guide rail, simplified arm-leg coupling system, and integrated resistance and rebound mechanisms ensure a biomechanically friendly, efficient, and enjoyable climbing experience. The flexibility to adapt materials, configurations, and features highlights the versatility and ingenuity of this invention.

Third Embodiment of the Exercise Device (**300**)

[0073] The third embodiment of the exercise device, referenced as **300** and illustrated in FIGS. **19-22**, introduces an open-frame guide column design, offering a more economical alternative to embodiments **200** and **100**. While maintaining the core functionality of the dynamic load transfer system **350** and “crawling” motion simulation, this embodiment achieves cost efficiency by simplifying structural components and reducing the number of rollers required for operation.

[0074] The open-frame guide column **310** replaces the enclosed guide rail column **210** of embodiment **200**, eliminating the need for pass-through slots **212**. This design allows for the inclusion of an additional set of rollers **343b**, positioned substantially below the upper roller set **343u**, as shown in FIG. **19**. The two roller sets, **343u** and **343b**, are fixed to the stationary frame structure **310** and **328**, providing guided rolling support for both the left and right arm poles **342**. These rollers allow the arm poles **342** to travel the full vertical range of approximately 21 inches, ensuring smooth operation during exercise.

[0075] The dual roller sets (**343u** and **343b**) offer significant structural stability, enabling a reduction in the number of guide rollers required for the leg carriages **330** compared to earlier embodiments **100** and **200**. In this design, the leg carriages **330** are lighter and less substantial, as their structural integrity is reinforced by their rigid coupling to the arm poles **342**, which serve as a stabilizing framework.

[0076] Additional support for the leg carriages **330** is provided by a lower rear-mounted set of guide rods **311**, best seen in FIG. **22**. These guide rods act as rails for rollers **395t**, mounted on the lower rear end of each leg carriage **330**. The rollers **395t** surround the front and back sides of the guide rods **311**, ensuring proper alignment and stability during operation. A forward-mounted roller **397** further maintains alignment by riding against the side wall of the forward frame structure **328**.

[0077] Similar to earlier embodiments, this design facilitates a “crawling” movement pattern, wherein the arm poles **342** cross over and connect rigidly to the opposite leg carriages **330**. For instance, the left arm pole is coupled to the right leg carriage and vice versa, allowing synchronized and efficient upper and lower body movements. This rigid coupling eliminates the need for cable systems used in embodiment **100**, further simplifying the mechanical design.

[0078] Embodiment **300** also incorporates features for improved portability and storage. A modified horizontal base member **322** includes travel rollers **327**, as depicted in FIGS. **21** and **22**. These travel rollers rotate around a round tube that pivots from brackets welded to each end of the base member **322**. This configuration allows the device to be stowed in a narrower footprint compared to earlier embodiments, enhancing its suitability for space-conscious users or environments.

[0079] The open-frame design of embodiment **300** provides opportunities for additional material substitutions, such as using linear bearings or bushings in place of the roller sets **343**, potentially further reducing manufacturing costs. Despite its simplified construction, this embodiment retains the robust functionality and fluid climbing motion of earlier designs while offering a more economical and compact solution.

Fourth Embodiment (**400**)

[0080] A fourth embodiment of the exercise device, designated as embodiment **400** and shown in FIGS. **23-26**, is now described. While visually and functionally similar to embodiment **200**, this embodiment incorporates distinct structural modifications. Specifically, the guide rail column **410** in embodiment **400** is constructed from a single rectangular metal tube, as opposed to the wooden or alternative material used for column **210** in embodiment **200**.

[0081] The guide rail column **410** features twin welded steel angle segments **411** on the backside and round rods **412** along the front side edges, providing vertical guidance for the roller carriages **430**. These guide features accommodate the rear V-rollers **495r** and forward U-groove rollers **495f** mounted to the leg carriage assemblies **430**, ensuring smooth operation and alignment.

[0082] Although earlier embodiments reference the guide rail column as being wooden, this material choice is illustrative only. The guide rail column may alternatively be constructed from materials such as metals (e.g., steel or aluminum), plastics, composites, or combinations thereof, depending on requirements for strength, weight, cost, aesthetics, and manufacturing feasibility.

[0083] The rebound system **480** in embodiment **400** introduces a single compression spring **483**, replacing the separate extension springs **283** used in the rebound system **280** of embodiment **200**. As illustrated in FIG. **26**, this compression spring **483** is housed inside a shortened square metal

tube **428** that also serves as a forward structural guide member.

[0084] The compression spring **483** interacts with a rebound rope **481** via an idler pulley **482**, providing a controlled rebound effect similar to earlier embodiments. The idler pulley **482** is mounted to a center pin fixed within a plastic slide block, which is confined within the square tube **428** but allowed to travel linearly. The spring **483**, positioned above the plastic slide block, applies downward force, as shown in the sectioned illustration of FIG. **26**, which omits certain components for clarity.

[0085] A milled slot **429** on the lower rearward sidewall of the square tube **428** permits the center pin of the pulley to extend outward. The spring tension may be adjusted using a threaded knob **487** positioned at the top of the square tube **428**. By turning the knob, the spring compression can be increased or decreased, allowing the user to modify the rebound intensity. Removing the knob entirely releases tension, enabling easy removal of the rebound rope, compression spring, or other internal components.

[0086] The roller carriages **430** and their guidance system are critical to the performance and user experience of the exercise device. In embodiment **400**, the carriages engage with the guide system through rear V-rollers **495r** and forward U-groove rollers **495f**. These rollers minimize friction while maintaining alignment during operation. Alternative configurations, such as angularly mounted rollers or linear bearings, may be implemented to further enhance performance.

[0087] To reduce noise and improve the quality of motion, the guide system and rollers may incorporate materials such as rubber, elastomers, or noise-dampening linings. These materials help eliminate side contact friction and produce a quieter, smoother experience for the user.

[0088] The flywheel and resistance system **450** in embodiment **400** is similar to those of earlier embodiments. The flywheel **455** is affixed to the transfer shaft **451** and is constructed from aluminum or another suitable material. Magnetic resistance is applied through a pair of adjustable magnets **453** that surround the flywheel. These magnets are adjusted using a pivoting, spring-loaded knob **457**, which includes a peg that can engage or disengage from a slotted selector disk **415**, as shown in FIG. **24**. Lowering the magnets increases resistance by enhancing the magnetic eddy current effect on the flywheel, while raising them reduces resistance. Additionally, the flywheel provides momentum to smooth out the climbing motion without hindering rapid directional changes.

[0089] The integration of the compression spring-based rebound system **480**, the refined roller guidance system, and the adjustable resistance mechanism further enhances the fluidity, efficiency, and overall performance of embodiment **400**. These features collectively improve the climbing experience while maintaining a compact and visually appealing design.

Fifth Embodiment (**500**)

[0090] Referring now to FIG. **27**, FIG. **27A**, and FIG. **28**, a fifth embodiment of the exercise device is designated by reference numeral **500**. As shown from a front-right perspective in FIG. **27**, the machine **500** includes an upright column **510** and a forward upper support tube **514**, within which a front kickstand **516** is pivotally mounted. The front kickstand **516** provides a fourth point of floor contact when in the operating position, stabilizing the device while maintaining a compact forward footprint.

[0091] The region enclosed by phantom lines labeled “A-A” in FIG. **27** highlights the area housing both the pivotal mount and the locking mechanism for the front kickstand **516**, as well as the centrally positioned flywheel **555**. FIG. **27A** provides a sectional view of these components, revealing internal structures otherwise obscured behind the upright column **510** and forward support tube **514**.

[0092] The top end of the front kickstand **516** features a short horizontal round tube with an offset bar **517** welded or otherwise affixed to its backside. At the opposite end of the bar **517**, a pivot hole engages and supports a generally U-shaped formed flat bar **518**. When the front kickstand **516** is pivoted inward into a stowed position, the U-shaped bar **518** moves upward, pressing against the

forward surface of the upright column **510**. To secure the kickstand in this stowed state, a clamping knob **519** with a threaded carriage bolt is provided. The bolt passes through the U-shaped bar **518**, and tightening the knob clamps the bar firmly against the column **510**. A milled slot in the column accommodates the travel of the offset bar **517** during this pivoting action, ensuring smooth operation and proper alignment.

[0093] Unlike earlier embodiments, the flywheel **555** in embodiment **500** is centrally located within the upper portion of the upright column **510**. This configuration necessitates front and rear slots in the column to accommodate the flywheel's central positioning. This arrangement not only offers aesthetic appeal but also simplifies the shrouding of the webbing spools located at opposite ends of the horizontal load transfer shaft. A single shroud **559** with a pass-through slot for the webbing effectively covers the spools, contributing to a cleaner and more streamlined appearance. Additionally, the central positioning of the flywheel **555** allows the adjustable magnetic resistance mechanism to pivot about the same axis that supports the two forward arm-pole guide rollers **543t**.

[0094] The adjustable resistance mechanism includes a pair of magnets **553** mounted on opposing sides of the aluminum flywheel **555**. The mechanism is adjusted using a spring-loaded pullout handle **557** connected to a selector plate with multiple indexed slots. By pulling the handle outward and tilting the mechanism to the desired angle, the user can vary the proximity of the magnets to the flywheel. When released, a locking peg engages one of the slots in the selector plate, securing the mechanism's position. The highest slot corresponds to maximum magnetic resistance, while the lowest slot positions the magnets away from the flywheel, providing minimal or no resistance.

[0095] Another feature introduced in embodiment **500** is a pair of short handles mounted near hip level on the backside of the upright column **510**, or at another convenient elevation. These handles assist users in mounting and dismounting the machine and can also be used during leg-only exercises when the arm poles are not engaged.

[0096] FIG. **28** illustrates a rear-left perspective of the machine **500** in its upright, stowed configuration. Both the front kickstand **516** and the rear kickstand are folded inward, enabling the device to stand more vertically and occupy minimal floor space. This stowed orientation facilitates easy rolling and allows the device to be conveniently stored against a wall or in a compact space when not in use.

[0097] From the above, it will be appreciated that a climbing exercise device according to an embodiment of the invention comprises a base configured to rest on a floor surface, an upright support column extending generally upward from the base, a first foot carriage and a second foot carriage, each carriage mounted for reciprocal movement along the upright support column such that, when one foot carriage moves upward, the other foot carriage moves downward, a horizontal transfer shaft rotatably supported on the upright support column at an elevation above a highest operational position of the foot carriages, and a load-transfer webbing system comprising at least two separate segments of flexible webbing, each segment having a first end attached to a respective one of the foot carriages and a second end attached to the horizontal transfer shaft in an opposing orientation, such that, when the first foot carriage moves downward and unwinds its webbing segment from the transfer shaft, the other webbing segment is wound onto the transfer shaft to lift the second foot carriage, wherein the load-transfer webbing system is configured to produce a varying load ratio on the first and second foot carriages during use so that resistance to downward motion is reduced when a foot carriage is at an upper position and increases as that foot carriage moves downward, thereby reducing knee strain at the start of the downward stroke.

[0098] In some embodiments, the climbing exercise device further comprises a rebound system operably coupled to the load-transfer webbing system, the rebound system including at least one spring, elastic band, or stretchable ligament arranged to maintain tension in each segment of flexible webbing throughout the full range of carriage travel and to progressively increase said tension as either foot carriage approaches an upper or lower limit of its stroke, thereby absorbing momentum at the travel extremes and facilitating a continuous, fluid climbing motion free from

slack or abrupt stops.

[0099] In some embodiments, the climbing exercise device further comprises a flywheel rigidly affixed to the horizontal transfer shaft so as to rotate in unison therewith, wherein the diameter of the transfer shaft or a spool thereon is sufficiently small that each full downward stroke of a foot carriage causes multiple rotations of the flywheel, thereby providing momentum to smooth the user's climbing motion without generating excessive inertia that would hinder quick directional changes enabled by the rebound system.

[0100] In some embodiments, the horizontal transfer shaft includes a spool having a sufficiently small diameter such that a full stroke of a foot carriage results in multiple rotations of the flywheel, thereby achieving a higher rotational velocity for a given foot carriage displacement, permitting a simpler, more economical magnetic resistance assembly with fewer parts, and providing adequate inertial smoothing without requiring large magnets, a large-diameter flywheel, or additional gearing to generate sufficient spin.

[0101] In some embodiments, the climbing exercise device further comprises a pair of arm handle assemblies, each assembly including an arm pole mounted for reciprocal movement along the upright support column; and a cross-member linkage disposed in a lower portion of the upright support column; wherein the left arm handle assembly is mechanically coupled to the right foot carriage and the right arm handle assembly is mechanically coupled to the left foot carriage, thereby facilitating a contra-lateral climbing motion while keeping upper portions of the arm handle assemblies free from overlapping linkages.

[0102] In some embodiments, a pair of arm handle assemblies, each assembly including an arm pole mounted for reciprocal movement along the upright support column; and at least one mating socket or bracket located in a lower portion of the upright support column; wherein each arm pole is selectively removable from its mating socket or bracket so that the contralateral linkage to the opposite foot carriage can be disengaged, and wherein said arm handle assemblies are likewise attachable and detachable in a version of the device configured for a standard climbing movement pattern, thereby enabling compact storage, transport, or shipping of the device without the arm handle assemblies attached.

[0103] In some embodiments, the climbing exercise device further comprises a stowable kickstand pivotally mounted to the base, the kickstand including a clamping assembly configured to frictionally engage the upright support column when in a folded or stowed position, and a locking knob adapted to secure the kickstand in either a folded position or an operational position, thereby permitting stable use of the device during operation and compact storage or transport when the kickstand is folded against the upright support column.

[0104] In some embodiments, the climbing exercise device includes the dynamic load transfer system, the rebound system, and the flywheel, wherein the dynamic load transfer system adjusts the load ratio across the user's limbs to reduce knee strain, the rebound system maintains continuous webbing tension and increases said tension at both upper and lower stroke limits to facilitate quick, fluid directional changes, and the flywheel's rotational inertia smooths transitions between upward and downward strokes, so that the combined interaction of these three features produces a climbing motion with reduced abrupt directional changes and lower knee stress compared to conventional climbers lacking such synergy.

Variations and Applications

[0105] The dynamic load transfer system described in the present invention is not limited to vertical climbing machines. Its underlying principles can be adapted for use in a variety of exercise equipment, including stair steppers and other devices simulating back and forth movement. By integrating this system, manufacturers can enhance user experience, reduce joint strain, and create more engaging workout routines that cater to a broader range of users. Furthermore, the magnetic braking system, as employed in embodiments **200** through **500**, offers a versatile, economical, and reliable method for providing adjustable resistance. This system can similarly be adapted for

alternative exercise devices to further enhance their functionality.

[0106] The load transfer between foot pedals or similar components may be accomplished through various mechanisms beyond the webbing-based system illustrated herein. Alternative flexible webbing segments include rope or cable mechanisms wrapped around drums or cam-shaped sheaves. Mechanical linkages such as levers or cams may also be used. These configurations can be tailored to dynamically distribute the user's weight and create varying resistance profiles, ensuring a comfortable and efficient exercise experience.

[0107] In a further embodiment, the exercise device may incorporate an electronic control system to regulate load transfer and resistance with precision. Such a system could include sensors, a processor, and actuators. Sensors monitor user inputs, such as force or speed, and device parameters, while the processor analyzes the data to determine appropriate resistance levels. Actuators, such as electric motors or generators, could then apply or resist the user's movement as necessary. This electronic control system would enable highly customizable resistance profiles and adaptable workout experiences, catering to diverse fitness goals and user preferences.

[0108] To optimize performance and user experience, the exercise device could combine multiple resistance generation methods. For example, a hydraulic system could be used alongside spring-based mechanisms or electronic controls to create a complex, adaptable resistance profile. Such a hybrid system would provide users with a highly responsive and versatile workout environment.

[0109] While the upright support column is illustrated herein as being straight, it will be appreciated that the column could be curved. It should also be appreciated that the single-column design described in the embodiments herein represents just one possible configuration. The device may be alternatively designed with multiple frame supports or guide rail columns to accommodate varying user preferences or design objectives. These alternative configurations maintain the core principles of dynamic load transfer and resistance generation, ensuring the same enhanced exercise experience as the single-column embodiments.

Claims

1. A climbing exercise device comprising: a base configured to rest on a floor surface; an upright support column extending upwardly from the base; right and left foot carriages, each of the foot carriages movable upwardly and downwardly along the upright support column; a horizontal transfer shaft rotatably supported on the upright support column at an elevation above a highest operational position of the foot carriages; and a load-transfer webbing system comprising two flexible webbing segments, each flexible webbing segment having a first end attached to a respective one of the foot carriages and a second end attached to the horizontal transfer shaft; wherein respective second ends of the flexible webbing segments are configured to wrap around the horizontal transfer shaft in opposite directions, such that, when one foot carriage moves upwardly and the other foot carriage moves downwardly, the flexible webbing segment attached to the upwardly moving foot carriage winds around the horizontal transfer shaft and the flexible webbing segment attached to the downwardly moving foot carriage unwinds from the horizontal transfer shaft.
2. The climbing exercise device of claim 1, further comprising a rebound system operably coupled to the load-transfer webbing system, the rebound system including an elastic cable assembly connecting the foot carriages via a pulley to maintain tension in each flexible webbing segment throughout a full range of foot carriage travel and to progressively increase tension as the foot carriages approach respective upper and lower limits of their stroke.
3. The climbing exercise device of claim 1, further comprising a flywheel rigidly affixed to the horizontal transfer shaft so as to rotate in unison therewith, wherein a diameter of the transfer shaft is sufficiently small that each full downward stroke of one of the foot carriages causes multiple rotations of the flywheel.

4. The climbing exercise device of claim 3, wherein the horizontal transfer shaft includes a spool having a sufficiently small diameter such that a full stroke of one of the foot carriages results in multiple rotations of the flywheel.
 5. The climbing exercise device of claim 3, further comprising an adjustable brake configured to act on the flywheel.
 6. The climbing exercise device of claim 1, further comprising an adjustable brake configured to act on the horizontal transfer shaft.
 7. The climbing exercise device of claim 1, further comprising: a pair of arm handle assemblies, each arm handle assembly including an arm pole mounted for reciprocal movement along the upright support column and a handle mounted on the arm pole; and a linkage disposed in a lower portion of the upright support column below a lowest operational position of the handles to operably couple each arm handle assembly with a respective one of the foot carriages on a same side of the device as the arm handle assembly or on an opposite side of the device as the arm handle assembly.
 8. The climbing exercise device of claim 5, further comprising: a pair of receptacles coupled with the linkage, each of the receptacles being configured to receive one of the arm handle assemblies.
 9. The climbing exercise device of claim 1, further comprising a stowable kickstand assembly including a kickstand pivotally mounted to the base, the kickstand being movable between a folded or stowed position against the upright support column and an operating position pivoted away from the upright support column.
 10. The climbing exercise device of claim 7, wherein the stowable kickstand assembly further comprises a clamping assembly configured to frictionally engage the upright support column when the kickstand is in a folded or stowed position.
 11. The climbing exercise device of claim 7, wherein the stowable kickstand assembly further comprises a locking knob adapted to secure the kickstand in at least one of the folded position and the operational position.
 12. A climbing exercise device comprising: a base configured to rest on a floor surface; an upright support column extending upwardly from the base; right and left foot carriages, each of the foot carriages movable upwardly and downwardly along the upright support column; a horizontal transfer shaft rotatably supported on the upright support column at an elevation above a highest operational position of the foot carriages; and a load-transfer webbing system comprising two flexible webbing segments, each flexible webbing segment having a first end attached to a respective one of the foot carriages and a second end attached to the horizontal transfer shaft, wherein respective second ends of the flexible webbing segments are configured to wrap around the horizontal transfer shaft in opposite directions, such that, when one foot carriage moves upwardly and the other foot carriage moves downwardly, the flexible webbing segment attached to the upwardly moving foot carriage winds around the horizontal transfer shaft and the flexible webbing segment attached to the downwardly moving foot carriage unwinds from the horizontal transfer shaft; a rebound system operably coupled to the load-transfer webbing system, the rebound system including an elastic cable assembly connecting the foot carriages via a pulley to maintain tension in each flexible webbing segment throughout a full range of foot carriage travel and to progressively increase tension as the foot carriages approach respective upper and lower limits of their stroke; and a flywheel rigidly affixed to the horizontal transfer shaft so as to rotate in unison therewith, wherein a diameter of the horizontal transfer shaft is sufficiently small that each full downward stroke of one of the foot carriages causes multiple rotations of the flywheel.
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